

AI YouTube Automation Platform — Full Industry-Level SaaS Project

PROJECT OVERVIEW

Build a complete industry-level AI SaaS platform called:

“AutoTube AI”

This platform should allow users to generate fully automated YouTube videos using AI.

The system must support:

- Script generation
- AI voiceover
- AI video editing
- AI thumbnails
- AI subtitles
- SEO optimization
- Auto upload to YouTube
- Multi-language support
- Faceless video generation
- Shorts/Reels generation
- Viral prediction system
- Team collaboration
- Analytics dashboard

The project must be:

- production-ready
 - scalable
 - modular
 - clean architecture
 - enterprise-level code quality
 - mobile responsive
 - secure
 - API-first
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CORE FEATURES

1. Authentication System

Features:

- Register/Login
- JWT Authentication
- Google OAuth
- Password reset
- Email verification
- User profile
- Subscription plans

Tech:

- FastAPI Auth
 - JWT
 - OAuth2
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2. AI Script Generator

User enters:

- topic
- niche
- target audience
- video duration

AI generates:

- Hook
- Main content
- CTA
- SEO keywords
- Video structure

Use:

- Llama 3
- Qwen
- Mistral

Features:

- Multi-language scripts
 - Urdu support
 - Long-form + Shorts
 - Tone selection
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3. AI Voiceover System

Convert scripts to voice.

Features:

- Male/Female voices
- Emotional tones
- Urdu/English/Hindi
- Speed control
- Voice cloning support

Use:

- XTTS
 - ElevenLabs API
 - Coqui TTS
-

4. AI Video Generation Engine

Automatically:

- select stock clips
- add transitions
- add subtitles
- sync audio
- add zoom effects
- add background music

Use:

- FFmpeg
- MoviePy
- OpenCV

Must support:

- landscape
 - portrait
 - YouTube Shorts
 - TikTok
 - Instagram Reels
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5. Thumbnail Generator

Generate viral thumbnails using AI.

Features:

- Text overlays
- Face enhancement
- CTR optimization
- AI suggestions

Use:

- Stable Diffusion
 - FLUX
-

6. Subtitle Generator

Features:

- Auto subtitles
- Animated captions
- Multi-language subtitles
- TikTok-style captions

Use:

- OpenAI Whisper
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7. SEO Optimization Engine

Generate:

- Titles
- Descriptions
- Hashtags
- Tags
- Keywords

Add:

- Viral score
 - SEO score
 - CTR prediction
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8. Auto Upload System

Integrate:

- YouTube API

Features:

- Scheduled uploads
 - Playlist support
 - Multi-channel support
 - Auto publishing
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9. Analytics Dashboard

Dashboard must show:

- Total videos
- Watch time
- CTR
- Audience retention
- SEO analytics
- AI recommendations

Charts required.

10. AI Shorts Generator

Convert long videos into:

- Shorts
- Reels
- TikTok videos

Features:

- Auto clipping
 - Auto captions
 - Face tracking
 - Viral moments detection
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11. Team Collaboration

Features:

- Multiple team members
 - Shared projects
 - Roles & permissions
 - Comments system
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12. SaaS Subscription System

Plans:

- Free
- Pro
- Enterprise

Integrate:

- Stripe
- PayPal

Features:

- Usage limits
- Credits system
- Billing dashboard

TECH STACK

FRONTEND

Use:

- Next.js
- React
- Tailwind CSS
- TypeScript
- Shadcn UI

Requirements:

- modern UI
- dark/light mode
- responsive dashboard
- clean animations

BACKEND

Use:

- FastAPI
- Python

Architecture:

- microservices
- async APIs
- modular architecture

Requirements:

- clean code
 - scalable
 - REST APIs
 - Swagger docs
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DATABASE

Use:

- PostgreSQL

Add:

- Redis caching
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STORAGE

Use:

- AWS S3

Store:

- videos
 - thumbnails
 - audio
 - subtitles
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VIDEO PROCESSING

Use:

- FFmpeg
 - OpenCV
 - MoviePy
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AI STACK

LLMs

- Llama 3
- Qwen
- Mistral

Speech

- XTTS
- Whisper

Image AI

- FLUX
 - Stable Diffusion
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DEVOPS

Use:

- Docker
- Docker Compose
- Nginx

Deployment:

- AWS
 - Railway
 - Render
 - Vercel
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SECURITY

Implement:

- rate limiting
 - API security
 - JWT refresh tokens
 - HTTPS
 - secure uploads
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DATABASE TABLES

Create tables for:

- users

- subscriptions
 - projects
 - scripts
 - videos
 - thumbnails
 - uploads
 - analytics
 - team_members
 - API_logs
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FOLDER STRUCTURE

Create professional folder structure for:

- frontend
 - backend
 - AI services
 - video engine
 - APIs
 - database
 - configs
 - utilities
 - Docker
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API REQUIREMENTS

Create APIs for:

- auth
 - script generation
 - voice generation
 - video rendering
 - thumbnail generation
 - subtitle generation
 - uploads
 - analytics
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UI PAGES

Create:

- landing page
 - pricing page
 - dashboard
 - video editor
 - analytics page
 - settings
 - subscription page
 - admin panel
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ADMIN PANEL

Features:

- manage users
 - monitor API usage
 - monitor rendering jobs
 - subscription management
 - analytics
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ADVANCED FEATURES

Add:

- AI agents
 - workflow automation
 - queue system
 - GPU task scheduling
 - webhook system
 - notifications
 - email system
 - AI recommendations
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BACKGROUND JOBS

Use:

- Celery
- Redis queues

For:

- video rendering
 - AI processing
 - uploads
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PROJECT REQUIREMENTS

Code must:

- be clean
- documented
- scalable
- modular
- production-ready

Include:

- README.md
 - setup guide
 - Docker setup
 - environment configs
 - API docs
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ADDITIONAL REQUIREMENTS

Create:

- sample datasets
 - sample prompts
 - demo AI workflows
 - testing scripts
 - CI/CD pipeline
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FINAL GOAL

Build a complete SaaS-level AI YouTube Automation Platform comparable to:

- Pictory
- InVideo
- VidIQ
- Opus Clip

But with:

- better AI automation
- Urdu support
- Shorts automation
- one-click workflow

Generate all required code files for frontend, backend, APIs, database, Docker, deployment, and AI integrations.

Use best practices everywhere.